the PCI-E lanes on the motherboard. Good move, we think, and long overdue.

It's not just about launching refreshed graphics cards but also working on tech that matters to the gaming fraternity, and to elucidate the point, AMD's Raja Koduri outlined three important pillars of the company's 'Radeon is Gaming' offerings. It's about AMD's GCN (Graphics Core Next) Architecture, TrueAudio technology, and Super high-resolution gaming. Most of the specifications improvement in the R9 and R7 series GPUs are all thanks to AMD's GCN Architecture, which also claims to provide enhanced DirectX 11.2 support and ramp up on energy efficiency. In its quest to be the definitive gaming platform out there, AMD's also taking the leap to fully support UltraHD 4K displays for enhanced visual brilliance – along with the multi monitor EyeFinity technology that AMD graphics cards support out of the box. With a Catalyst Driver update, the UltraHD 4K display support will also be rolled out to current GCN enabled AMD graphics cards.

AMD TrueAudio Technology

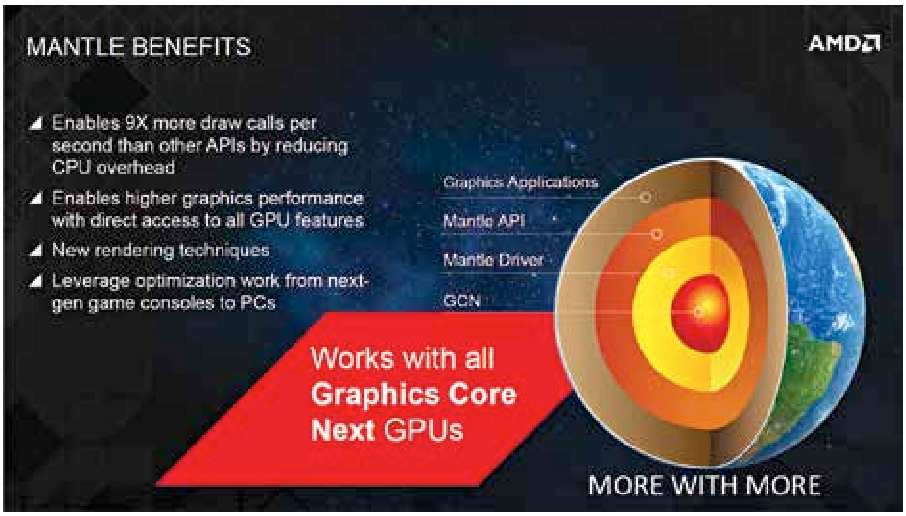
Another integral part of the new AMD Radeon graphics offering is the company's TrueAudio Technology. With a lot of impetus given on enhancing visual, 3D graphics as far as GPUs are concerned, it's about time that someone did something about optimizing and enhancing the audio experience while gaming – a key component for true in-game immersion. What AMD tries to deliver with


Radeon R9-290X: AMD's new flagship

TrueAudio is a fully programmable audio engine for game developers to work with. While AMD hopes to "revolutionize game audio" with TrueAudio Technology – we'll just have to wait and watch – a game that exploits this feature of the DSP will be able to reproduce more realtime voices and channels, letting you perceive a greater degree of directional audio over any output. Well, recreating complex positional sound with audio functions isn't something new, what is new is AMD's approach to make the whole process independent of the main CPU (to a large extent). The company claims that only 10 per cent of a game's CPU resources are traditionally allocated for audio processing, and with TrueAudio it's trying to make real-time CPU-independent audio processing available for games to exploit.

According to AudioKinetic – they design audio middleware – traditionally it's either the CPU or a dedicated DSP (assisted by the CPU) that processes in-game sound on the fly, but that isn't the case with AMD TrueAudio. Digit was present at the event listening to AMD's partner's sound demos across 24 channels squeezed into a 7.1 setup, and the directional and positional sound was definitely impressive; even on a stereo demo, the same audio clip didn't deteriorate overwhelmingly, retaining a significant amount of the directional audio processed on-the-fly on a Radeon R9 series GPU.

This new TrueAudio technology will be made available on AMD Radeon R9 290X, R9 290 and R7 260X GPUs only (for now). All three GPUs claim to have three dedicated DSP cores for audio processing in their attempt to deliver enhanced audio immersiveness. Upcoming games like *Thief*, *Murdered Soul Suspect*, *Lichdom*, *Sonic & All Stars Racing Transformed*, and of course *Battlefield 4* are all developed to exploit AMD's TrueAudio tech, if you're gaming on a supported Radeon GPU, obviously. We can't really comment on audio performance conclusively without testing one of these TrueAudio enabled GPUs, but all signs indicate that the era of superior in-game audio and audiophile gamers is about to take off real quick. That's got to be good for gamers, developers, and the industry as a whole, right?



Project Mantle claims to offer developers unrestricted access to the GPU's full potential

Project Mantle

This is AMD's silver bullet. Mantle is AMD's solution to something that developers have been "complaining about" for a long time now – flip to Page 113 to know more about this rant. It's a low-level API in its GCN architecture that's different from DirectX, Direct3D, OpenGL, etc., allowing game engines to knock on the GPU's hardware directly – demand GPU memory access – reduce CPU draws and boost overall performance. More importantly, with AMD's APUs (and GCN) debuting on the upcoming Xbox One and PS4 consoles (and a lot of Radeon GPUs down the line), it offers game developers a one-stop solution to code and port their game engines smoothly across platforms and devices. This definitely throws open interesting possibilities for PC gaming, and we'll be watching AMD closely to see how it all unfolds in the near future.